Guardrails

Guardrails are safety mechanism that control what goes inot and comes out of an ai agent.

They Sit around your agent pipeline and ensure the agent.

There are two approaches to use guardrails

1. Deterministic Guardrails

These use **fixed rules and logic** instead of another AI model.

They are predictable because the output is based on predefined conditions.

How They Work

- Regex
- Keyword matching
- JSON schema validation
- Rule engines
- If-else logic
- Type checking

2. Model-Based Guardrails

These use another **LLM or classifier model** to evaluate inputs/outputs.

Instead of hardcoded rules, the guardrail model “understands” the content semantically.

<u>LangChain implements guardrails in the form of middleware</u>

1)PII Middleware

PII Middleware is a guardrail layer that detects, masks, blocks, or protects **Personally Identifiable Information (PII)** in AI applications.

PII means sensitive personal data like:

- phone numbers
- emails

- Aadhaar numbers
- passport numbers
- credit card details
- addresses
- passwords

2)Human Loop middleware

Human-in-the-Loop (HITL) Middleware is a guardrail layer where a **human reviews, approves, edits, or rejects** an AI system's action before it continues.

It is used when AI decisions are:

- high risk
- expensive
- security-sensitive
- legally important

3)Before Agent Hook

"Before agent hook" usually refers to a **middleware/interceptor step that runs *before an AI agent starts executing*** its reasoning, tool calls, or workflow.

It's basically a **pre-processing guardrail for agents**.

4)After Agent Hook

->Validates final response before user sees it

->Replace or mutat unsafe content

5)Layered Guardrails

Layered guardrails means using **multiple safety and validation layers stacked together** in an AI system, instead of relying on just one check.

The idea is simple:

Don't trust a single safeguard → protect the system at multiple stages.